Data Cleaning

In [112…
```
import pandas as pd
```

In [113…
```
#loading sales data
df= pd.read_excel('//Sales_data.xlsx')
```

In [114…
```
#Basic checks
print(df.shape)
df.head()
df.info()
df.columns
```

```
(51290, 21)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 51290 entries, 0 to 51289
Data columns (total 21 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   order_id        51290 non-null  object
 1   order_date      51290 non-null  datetime64[ns]
 2   ship_date       51290 non-null  datetime64[ns]
 3   ship_mode       51290 non-null  object
 4   customer_name   51290 non-null  object
 5   segment         51290 non-null  object
 6   state           51290 non-null  object
 7   country         51290 non-null  object
 8   market          51290 non-null  object
 9   region          51290 non-null  object
 10  product_id      51290 non-null  object
 11  category        51290 non-null  object
 12  sub_category    51290 non-null  object
 13  product_name    51290 non-null  object
 14  sales           51290 non-null  int64
 15  quantity        51290 non-null  int64
 16  discount        51290 non-null  float64
 17  profit          51290 non-null  float64
 18  shipping_cost   51290 non-null  float64
 19  order_priority  51290 non-null  object
 20  year            51290 non-null  int64
dtypes: datetime64[ns](2), float64(3), int64(3), object(13)
memory usage: 8.2+ MB
```

Out[114…
```
Index(['order_id', 'order_date', 'ship_date', 'ship_mode', 'customer_name',
       'segment', 'state', 'country', 'market', 'region', 'product_id',
       'category', 'sub_category', 'product_name', 'sales', 'quantity',
       'discount', 'profit', 'shipping_cost', 'order_priority', 'year'],
      dtype='object')
```

In [115…
```
#checking nulls

df.isna().sum()

#drop nulls if any
```

```
df.dropna()
```

Out[115…

	order_id	order_date	ship_date	ship_mode	customer_name	segment	state
0	AG-2011-2040	2022-01-01	2022-01-06	Standard Class	Toby Braunhardt	Consumer	Constantine
1	IN-2011-47883	2022-01-01	2022-01-08	Standard Class	Joseph Holt	Consumer	New South Wales
2	HU-2011-1220	2022-01-01	2022-01-05	Second Class	Annie Thurman	Consumer	Budapest
3	IT-2011-3647632	2022-01-01	2022-01-05	Second Class	Eugene Moren	Home Office	Stockholm
4	IN-2011-47883	2022-01-01	2022-01-08	Standard Class	Joseph Holt	Consumer	New South Wales
...	...	...	...	...	...	...	...
51285	CA-2014-115427	2025-12-31	2026-01-04	Standard Class	Erica Bern	Corporate	California
51286	MO-2014-2560	2025-12-31	2026-01-05	Standard Class	Liz Preis	Consumer	Souss-Massa-Draâ
51287	MX-2014-110527	2025-12-31	2026-01-02	Second Class	Charlotte Melton	Consumer	Managua
51288	MX-2014-114783	2025-12-31	2026-01-06	Standard Class	Tamara Dahlen	Consumer	Chihuahua
51289	CA-2014-156720	2025-12-31	2026-01-04	Standard Class	Jill Matthias	Consumer	Colorado

51290 rows × 21 columns

In [116…

```
#create new column month_year

df['Month_Year']= df['order_date'].dt.strftime('%m-%Y')
df['Month_Year']
```

Out[116…

```
0        01-2022
1        01-2022
2        01-2022
3        01-2022
4        01-2022
          ...
51285    12-2025
51286    12-2025
51287    12-2025
51288    12-2025
51289    12-2025
Name: Month_Year, Length: 51290, dtype: object
```

In [120…

```
# Customize Market column by considering only 4 regions(APAC, EMEA, LATAM, NA), res

for i in df.market:
    if i in ('Africa', 'EU'):
        df['market']=df['market'].replace(i, 'EMEA')
    if i in ('Canada','US'):
        df['market']= df['market'].replace(i, 'NA')
df['market']
```

Out[120…

```
0         EMEA
1         APAC
2         EMEA
3         EMEA
4         APAC
         ...
51285       NA
51286     EMEA
51287    LATAM
51288    LATAM
51289       NA
Name: market, Length: 51290, dtype: object
```

In [121…

```
# Removing extraneous info from Product name column
df['product_name']= df['product_name'].str.split(',').str[0]
df['product_name']
```

Out[121…

```
0                          Tenex Lockers
1                           Acme Trimmer
2                              Tenex Box
3                      Enermax Note Cards
4                        Eldon Light Bulb
                       ...
51285        Cardinal Slant-D Ring Binder
51286    Wilson Jones Hole Reinforcements
51287              Hon Color Coded Labels
51288            Hon Legal Exhibit Labels
51289                 Bagged Rubber Bands
Name: product_name, Length: 51290, dtype: object
```

In [122…

```
df.columns
```

Out[122…

```
Index(['order_id', 'order_date', 'ship_date', 'ship_mode', 'customer_name',
       'segment', 'state', 'country', 'market', 'region', 'product_id',
       'category', 'sub_category', 'product_name', 'sales', 'quantity',
       'discount', 'profit', 'shipping_cost', 'order_priority', 'year',
       'Month_Year'],
      dtype='object')
```

In [125…

```
df['order_date']
```

Out[125…

```
0       2022-01-01
1       2022-01-01
2       2022-01-01
3       2022-01-01
4       2022-01-01
           ...
51285   2025-12-31
51286   2025-12-31
51287   2025-12-31
51288   2025-12-31
51289   2025-12-31
Name: order_date, Length: 51290, dtype: datetime64[ns]
```

In [133…

```
#Filtering only last 2 years data using loop

filtered_df= []
#for index, rows in df.iterrows():
 #   if rows['order_date'].year== 2024 or rows['order_date'].year==2025:
  #      filtered_df.append(rows)
#filtered_df= pd.DataFrame(filtered_df)

#Filtering only last 2 years data using vectorized method

filtered_df= df[(df['order_date'].dt.year==2024) | (df['order_date'].dt.year==2025)
filtered_df
```

Out[133…

	order_id	order_date	ship_date	ship_mode	customer_name	segment	state
19960	ID-2013-37383	2024-01-01	2024-01-04	First Class	Muhammed Yedwab	Corporate	New South Wales
19961	IN-2013-47960	2024-01-01	2024-01-06	Standard Class	Joe Elijah	Consumer	Tianjin
19962	ID-2013-37383	2024-01-01	2024-01-04	First Class	Muhammed Yedwab	Corporate	New South Wales
19963	IN-2013-47960	2024-01-01	2024-01-06	Standard Class	Joe Elijah	Consumer	Tianjin
19964	IN-2013-47960	2024-01-01	2024-01-06	Standard Class	Joe Elijah	Consumer	Tianjin
...	...	...	...	...	...	...	...
51285	CA-2014-115427	2025-12-31	2026-01-04	Standard Class	Erica Bern	Corporate	California
51286	MO-2014-2560	2025-12-31	2026-01-05	Standard Class	Liz Preis	Consumer	Souss-Massa-Draâ
51287	MX-2014-110527	2025-12-31	2026-01-02	Second Class	Charlotte Melton	Consumer	Managua
51288	MX-2014-114783	2025-12-31	2026-01-06	Standard Class	Tamara Dahlen	Consumer	Chihuahua
51289	CA-2014-156720	2025-12-31	2026-01-04	Standard Class	Jill Matthias	Consumer	Colorado

31330 rows × 23 columns

In [134…

```
#Formatting order_id column

df['new_order_id']= df['country'].str[:2].str.upper()+''+df['Month_Year'].str[2:]+'
filtered_df['order_id']= df['new_order_id']
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_10348\1665849063.py:4: SettingWithCopyWarni
ng:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/u
ser_guide/indexing.html#returning-a-view-versus-a-copy
  filtered_df['order_id']= df['new_order_id']
```

In [135…

```
filtered_df.columns.str.upper()
```

Out[135…

```
Index(['ORDER_ID', 'ORDER_DATE', 'SHIP_DATE', 'SHIP_MODE', 'CUSTOMER_NAME',
       'SEGMENT', 'STATE', 'COUNTRY', 'MARKET', 'REGION', 'PRODUCT_ID',
       'CATEGORY', 'SUB_CATEGORY', 'PRODUCT_NAME', 'SALES', 'QUANTITY',
       'DISCOUNT', 'PROFIT', 'SHIPPING_COST', 'ORDER_PRIORITY', 'YEAR',
       'MONTH_YEAR', 'NEW_ORDER_ID'],
      dtype='object')
```

In [136…

```
# dropping unnecessary columns

filtered_df.drop(['region','year'], axis=1, inplace= True)
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_10348\403502480.py:3: SettingWithCopyWarnin
g:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/u
ser_guide/indexing.html#returning-a-view-versus-a-copy
  filtered_df.drop(['region','year'], axis=1, inplace= True)
```

In [137…

```
#renaming market to region

filtered_df.rename(columns= {'market':'region'}, inplace= True)
filtered_df
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_10348\3704880646.py:3: SettingWithCopyWarni
ng:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/u
ser_guide/indexing.html#returning-a-view-versus-a-copy
  filtered_df.rename(columns= {'market':'region'}, inplace= True)
```

Out[137…

	order_id	order_date	ship_date	ship_mode	customer_name	segment	state
19960	AU-2024-37383	2024-01-01	2024-01-04	First Class	Muhammed Yedwab	Corporate	New South Wales
19961	CH-2024-47960	2024-01-01	2024-01-06	Standard Class	Joe Elijah	Consumer	Tianjin
19962	AU-2024-37383	2024-01-01	2024-01-04	First Class	Muhammed Yedwab	Corporate	New South Wales
19963	CH-2024-47960	2024-01-01	2024-01-06	Standard Class	Joe Elijah	Consumer	Tianjin
19964	CH-2024-47960	2024-01-01	2024-01-06	Standard Class	Joe Elijah	Consumer	Tianjin
...	...	...	...	...	...	...	...
51285	UN-2025-115427	2025-12-31	2026-01-04	Standard Class	Erica Bern	Corporate	California
51286	MO-2025-2560	2025-12-31	2026-01-05	Standard Class	Liz Preis	Consumer	Souss-Massa-Draâ
51287	NI-2025-110527	2025-12-31	2026-01-02	Second Class	Charlotte Melton	Consumer	Managua
51288	ME-2025-114783	2025-12-31	2026-01-06	Standard Class	Tamara Dahlen	Consumer	Chihuahua
51289	UN-2025-156720	2025-12-31	2026-01-04	Standard Class	Jill Matthias	Consumer	Colorado

31330 rows × 21 columns

In [131…

```
# cleaned data set
filtered_df
```

Out[131…

	order_id	order_date	ship_date	ship_mode	customer_name	segment	state
19960	AU-2024-37383	2024-01-01	2024-01-04	First Class	Muhammed Yedwab	Corporate	New South Wales
19961	CH-2024-47960	2024-01-01	2024-01-06	Standard Class	Joe Elijah	Consumer	Tianjin
19962	AU-2024-37383	2024-01-01	2024-01-04	First Class	Muhammed Yedwab	Corporate	New South Wales
19963	CH-2024-47960	2024-01-01	2024-01-06	Standard Class	Joe Elijah	Consumer	Tianjin
19964	CH-2024-47960	2024-01-01	2024-01-06	Standard Class	Joe Elijah	Consumer	Tianjin
...	...	...	...	...	...	...	...
51285	UN-2025-115427	2025-12-31	2026-01-04	Standard Class	Erica Bern	Corporate	California
51286	MO-2025-2560	2025-12-31	2026-01-05	Standard Class	Liz Preis	Consumer	Souss-Massa-Draâ
51287	NI-2025-110527	2025-12-31	2026-01-02	Second Class	Charlotte Melton	Consumer	Managua
51288	ME-2025-114783	2025-12-31	2026-01-06	Standard Class	Tamara Dahlen	Consumer	Chihuahua
51289	UN-2025-156720	2025-12-31	2026-01-04	Standard Class	Jill Matthias	Consumer	Colorado

31330 rows × 20 columns

In [132…

```
#save cleaned dataset

filtered_df.to_excel('//cleaned_sales_data.xlsx', index= False)
```

In [ ]: